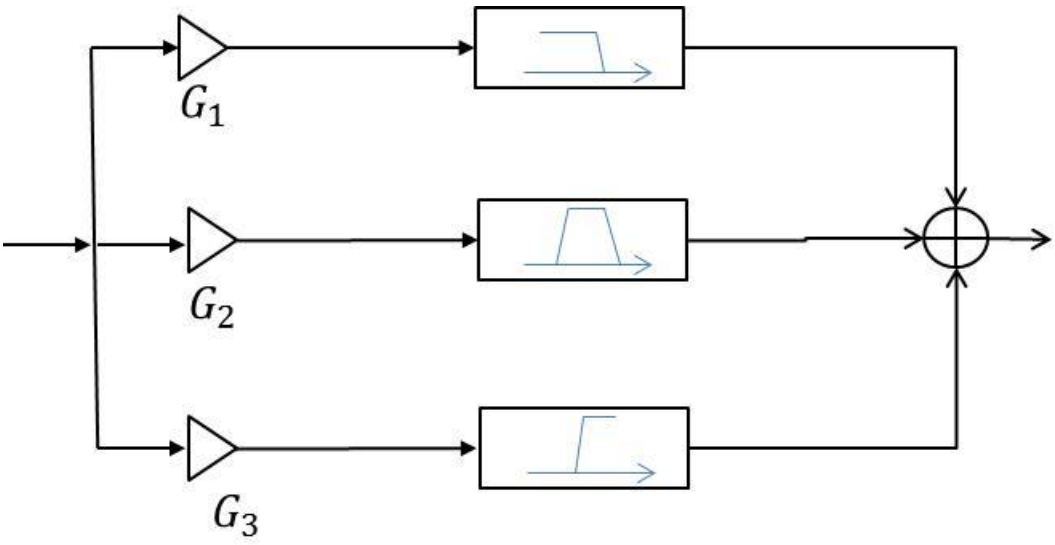
DSP Lab 11 Graphical Music Equalizer

Introduction

A music equalizer can adjust the amplitude of the audio signal at different frequency bands. This project will implement a 3-band music equalizer with the TI TMS320C6713DSK. The audio spectrum will be divided into 3 bands: 20 Hz – 200 Hz, 200 Hz – 2 KHz, 2 KHz – 20 KHz, each with adjustable gains. The block diagram of the 3-band equalizer is shown as follows, where $G_n \in [0,1]$ is the gain of the *n*-th band.



Pre-lab

1. Use fdatool in Matlab to design the three filters (low pass filter, band-pass filter, and high pass filter). Use FIR filter with Kaiser window, beta = 0.5, order = 80, Fs = 44100 Hz. Convert the filter parameters into coefficient files that can be loaded with the C program.

Lab Assignment

1. Implement the equalizer with Matlab. Adjust the gains at different bands and listen to the effects. (IMPORTANT: to avoid clipping, please divide the gain of the original audio signal by 3 before passing it through the equalizer).
2. Implement the equalizer in the DSK. Using the GEL slider (refer to lab 1) to control the gain. List to the different effects. (IMPORTANT: appropriate measures need to be taken to avoid clipping).